



భారతీయ సాంకేతిక విజ్ఞాన సంస్థ హైదరాబాద్
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Indian Institute of Technology Hyderabad

Mathematical Foundations of Data Science (Probability)

Lecture 07

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General Transformations

Fix $(\Omega, \mathscr{F}, \mathbb{P})$.
Let $X : \Omega \to \mathbb{R}$ be a **continuous random variable** with PDF f_X.

Suppose that we are given $g : \mathbb{R} \to \mathbb{R}$ that:

1. Is **(strictly) monotone and differentiable** throughout its domain.
2. Has **non-zero derivative everywhere** in its domain, i.e., $g'(x) \neq 0$ for all $x \in \mathbb{R}$.

What is the PDF of $Y = g(X)$?

Note that because g is strictly monotone, admits an inverse, say $g^{-1} : \mathbb{R} \to \mathbb{R}$

Case 1: g Strictly Increasing

$$
\begin{aligned}
F_Y(y) &= \mathbb{P}(\{Y \le y\}) \\
&= \mathbb{P}(\{g(X) \le y\}) \\
&= \mathbb{P}(\{X \le g^{-1}(y)\}) \\
&= \int_{-\infty}^{g^{-1}(y)} f_X(u)\,\mathrm{d}u \\
&\qquad \text{make substitution } g(u) = v \\
&= \int_{-\infty}^{y} \frac{f_X(g^{-1}(v))}{g'(g^{-1}(v))}\,\mathrm{d}v.
\end{aligned}
$$

Thus, we have

$$
f_Y(y) = \begin{cases} \dfrac{f_X(g^{-1}(y))}{g'(g^{-1}(y))}, & y \in \mathrm{Range}(g), \\ 0, & y \notin \mathrm{Range}(g). \end{cases}
$$

Case 2: g Strictly Decreasing

$$
\begin{aligned}
F_Y(y) &= \mathbb{P}(\{Y \leq y\}) \\
&= \mathbb{P}(\{g(X) \leq y\}) \\
&= \mathbb{P}(\{X \geq g^{-1}(y)\}) \\
&= \int_{g^{-1}(y)}^{\infty} f_X(u)\,\mathrm{d}u \\
&= \int_{y}^{\infty} \frac{f_X(g^{-1}(v))}{g'(g^{-1}(v))}\,\mathrm{d}v \qquad \text{make substitution } g(u) = v.
\end{aligned}
$$

It thus follows that

$$
f_Y(y) = \begin{cases} \dfrac{f_X(g^{-1}(y))}{-g'(g^{-1}(y))}, & y \in \mathrm{Range}(g), \\ 0, & y \notin \mathrm{Range}(g). \end{cases}
$$

General Transformations:

g Monotone, Differentiable

Jacobi's Transformation Formula

When $g : \mathbb{R} \to \mathbb{R}$ is **monotone and differentiable throughout its domain**,

$$Y = g(X), \qquad f_Y(y) = \begin{cases} \dfrac{f_X(g^{-1}(y))}{\left|g'(g^{-1}(y))\right|}, & y \in \text{Range}(g), \\ 0, & y \notin \text{Range}(g). \end{cases}$$

Example

Jacobi's Transformation Formula

Let $X \sim \mathcal{N}(0,1)$. Derive the PDF of $Y = e^X$.

- Here: $g(x) = e^x, \qquad \text{Domain}(g) = \mathbb{R}, \qquad \text{Range}(g) = (0, +\infty)$
- g monotone increasing and differentiable with non-zero derivative **throughout its domain**
- For any $y \in \text{Range}(g)$ (i.e., $y > 0$)

$$
\begin{aligned}
f_Y(y) &= \frac{f_X(g^{-1}(y))}{g'(g^{-1}(y))}, \qquad g^{-1}(y) = \ln y \\
&= \frac{f_X(\ln y)}{g'(\ln y)} \\
&= \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{(\ln y)^2}{2}}}{e^{\ln y}} = \frac{1}{y\sqrt{2\pi}} e^{-\frac{(\ln y)^2}{2}}, \qquad y > 0.
\end{aligned}
$$

(Log-normal distribution)

General Transformations: g Piecewise Monotone, Differentiable

When $g : \mathbb{R} \to \mathbb{R}$ is piecewise monotone and differentiable

Suppose that $I_1, \ldots, I_n$ is a partition of $\mathbb{R}$.
Suppose that $g : \mathbb{R} \to \mathbb{R}$ is **monotone** and **differentiable** on I_ℓ for each $\ell \in \{1, \ldots, n\}$.
Let $g(I_\ell)$ denote the image of I_ℓ under g
Let h_ℓ denote the inverse of g on I_ℓ.
If X is a continuous random variable with PDF f_X, then the PDF of $Y = g(X)$ is given by

$$f_Y(y) \quad = \quad \sum_{\ell=1}^{n} \frac{f_X\Big(h_\ell(y)\Big)}{\Big|g'\Big(h_\ell(y)\Big)\Big|} \mathbf{1}_{g(I_\ell)}(y).$$

Example

Suppose $X \sim \mathcal{N}(0, 1)$. Derive the PDF of $Y = X^2$.

- Here, $g(x) = x^2, \qquad \text{Domain}(g) = \mathbb{R}, \qquad \text{Range}(g) = [0, +\infty)$
- g is **differentiable** throughout its domain, and $g'(x) = 2x$ for all $x \in \mathbb{R}$
- g is neither monotone increasing nor monotone decreasing throughout its domain
- Partition $\mathbb{R}$ into $I_1 = (-\infty, 0), \quad I_2 = \{0\}, \quad I_3 = (0, +\infty)$
- g is **strictly decreasing on I_1** and **strictly increasing on I_3**, $\quad h_1(y) = -\sqrt{y}, \quad h_2(y) = \sqrt{y}$
- $g(I_1) = (0, +\infty), \quad g(I_2) = \{0\}, \quad g(I_3) = (0, +\infty)$
- For any $y \in \text{Range}(g)$,

$$
\begin{aligned}
f_Y(y) &= \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}}}{\left|-2\sqrt{y}\right|} \mathbf{1}_{(0,\,+\infty)}(y) + \frac{\frac{1}{\sqrt{2\pi}} e^{-\frac{y}{2}}}{\left|2\sqrt{y}\right|} \mathbf{1}_{(0,\,+\infty)}(y) \\
&= \frac{1}{\sqrt{2\pi y}} e^{-\frac{y}{2}} \mathbf{1}_{(0,\,+\infty)}(y) \qquad \text{(Chi-Squared distribution with 1 degree of freedom)}.
\end{aligned}
$$

General Transformations: Multivariate Case

- Suppose that we are given n "nice"[1] functions

$$g_1 : \mathbb{R}^n \to \mathbb{R}, \qquad g_2 : \mathbb{R}^n \to \mathbb{R}, \qquad \cdots \qquad g_n : \mathbb{R}^n \to \mathbb{R}.$$

- Define $g : \mathbb{R}^n \to \mathbb{R}^n$ as

$$g(x_1, \ldots, x_n) \;=\; \begin{pmatrix} g_1(x_1, \ldots, x_n) \\ \vdots \\ g_n(x_1, \ldots, x_n) \end{pmatrix}$$

General Transformations (Multivariate Case)

Suppose that $X_1, \ldots, X_n$ are jointly continuous random variables with joint PDF $f_{X_1,\ldots,X_n}$.
Let $Y_1, \ldots, Y_n$ be defined as

$$Y_\ell \;:=\; g_\ell(X_1, \ldots, X_n), \qquad \ell \in \{1, \ldots, n\}.$$

What is the joint PDF of $Y_1, \ldots, Y_n$ in terms of $f_{X_1,\ldots,X_n}$?

[1] We will assume that $g_1, \ldots, g_n$ are differentiable with continuous first-order partial derivatives.

Jacobian Matrix

Definition (Jacobian Matrix)

Let $g : \mathbb{R}^n \to \mathbb{R}^n$ be defined by

$$g(x_1, \ldots, x_n) = \begin{pmatrix} g_1(x_1, \ldots, x_n) \\ \vdots \\ g_n(x_1, \ldots, x_n) \end{pmatrix}.$$

The **Jacobian matrix** of g at the point $\mathbf{x} = (x_1, \ldots, x_n)$ is defined as

$$J_g(\mathbf{x}) = J_g(x_1, \ldots, x_n) = \begin{pmatrix} \frac{\partial g_1}{\partial x_1}(x_1, \ldots, x_n) & \cdots & \frac{\partial g_1}{\partial x_n}(x_1, \ldots, x_n) \\ \vdots & & \vdots \\ \frac{\partial g_n}{\partial x_1}(x_1, \ldots, x_n) & \cdots & \frac{\partial g_n}{\partial x_n}(x_1, \ldots, x_n) \end{pmatrix}$$

- $\det(J_g(x_1, \ldots, x_n))$ is referred to as the **Jacobian** of g at the point $\mathbf{x} = (x_1, \ldots, x_n)$

Jacobi's Transformation Formula

Let $X_1, \ldots, X_n$ be jointly continuous with joint PDF $f_{X_1,\ldots,X_n}$.

Jacobi's Transformation Formula

Let $g : \mathbb{R}^n \to \mathbb{R}^n$ be **one-one**, **differentiable** with **continuous first-order partial derivatives**, and **non-zero Jacobian throughout its domain**.
Let the individual components of g be denoted by $g_1, \ldots, g_n$.
Let $Y_\ell = g_\ell(X_1, \ldots, X_n), \quad \ell \in \{1, \ldots, n\}$.
Then, the joint PDF of $Y_1, \ldots, Y_n$ is given by

$$f_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n) = \begin{cases} \dfrac{f_{X_1,\ldots,X_n}\left(g^{-1}(y_1, \ldots, y_n)\right)}{\left|\det J_g\left(g^{-1}(y_1, \ldots, y_n)\right)\right|}, & (y_1, \ldots, y_n) \in \text{Range}(g), \\ 0, & (y_1, \ldots, y_n) \notin \text{Range}(g). \end{cases}$$

Example

Let $X_1, X_2 \overset{\text{IID}}{\sim} \text{Exponential}(\lambda)$ for some fixed $\lambda > 0$. Derive the joint PDF of $Y_1 = X_1$ and $Y_2 = X_1 + X_2$. Also deduce the conditional PDF of Y_1, conditioned on the event $\{Y_2 = y_2\}$.

- $g_1(x_1, x_2) = x_1, \qquad g_2(x_1, x_2) = x_1 + x_2,$
  $\text{Domain}(g_1) = [0, +\infty) \times [0, +\infty) = \text{Domain}(g_2), \qquad \text{Range}(g_1) = [0, +\infty) = \text{Range}(g_2)$
- $g : \mathbb{R}^2 \to \mathbb{R}^2$ and its Jacobian:

$$g(x_1, x_2) = \begin{pmatrix} g_1(x_1, x_2) \\ g_2(x_1, x_2) \end{pmatrix} = \begin{pmatrix} x_1 \\ x_1 + x_2 \end{pmatrix}, \qquad J_g(x_1, x_2) = \begin{pmatrix} \dfrac{\partial g_1}{\partial x_1}(x_1, x_2) & \dfrac{\partial g_1}{\partial x_2}(x_1, x_2) \\ \dfrac{\partial g_2}{\partial x_1}(x_1, x_2) & \dfrac{\partial g_2}{\partial x_2}(x_1, x_2) \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}$$

- $\text{Domain}(g) = [0, +\infty) \times [0, +\infty), \qquad \text{Range}(g) = \{(x, y) : 0 \le x \le y < +\infty\}$
- $\left| \det J_g(x_1, x_2) \right| = 1 \qquad \forall (x_1, x_2) \in \text{Domain}(g)$
- $g^{-1}(y_1, y_2) = \begin{pmatrix} y_1 \\ y_2 - y_1 \end{pmatrix} \qquad \forall (y_1, y_2) \in \text{Range}(g)$

Example

- For any $(y_1, y_2) \in \text{Range}(g)$,

$$\begin{aligned} f_{Y_1,Y_2}(y_1, y_2) &= \frac{f_{X_1,X_2}\left(g^{-1}(y_1, y_2)\right)}{\left|\det J_g\left(g^{-1}(y_1, y_2)\right)\right|} \\ &= f_{X_1,X_2}\left(y_1,\ y_2 - y_1\right) \\ &= f_{X_1}(y_1) \cdot f_{X_2}(y_2 - y_1) \\ &= \lambda\, e^{-\lambda y_1} \cdot \lambda\, e^{-\lambda(y_2 - y_1)} \\ &= \lambda^2\, e^{-\lambda y_2}, \qquad 0 \le y_1 \le y_2 < +\infty. \end{aligned}$$

- Conditional PDF:

$$f_{Y_1|\{Y_2=y_2\}}(y_1) = \frac{f_{Y_1,Y_2}(y_1, y_2)}{f_{Y_2}(y_2)} = \frac{\lambda^2\, e^{-\lambda y_2}}{\int_0^{y_2} \lambda^2\, e^{-\lambda y_2}\, \mathrm{d}y_1} = \frac{1}{y_2}, \qquad y_1 \in [0, y_2].$$

Jacobi's Transformation Formula: g Piecewise Differentiable with Non-Zero Jacobian

When $g : \mathbb{R}^n \to \mathbb{R}^n$ is piecewise differentiable with non-zero Jacobian

Suppose that $I_1, \ldots, I_n$ is a partition of $\mathbb{R}^n$, and $g : \mathbb{R}^n \to \mathbb{R}^n$ is **one-one**, **differentiable** with **continuous first-order partial derivatives**, and has non-zero Jacobian on I_ℓ for each $\ell \in \{1, \ldots, n\}$.
Let h_ℓ denote the inverse of g on I_ℓ.
Let the individual components of g be $g_1, \ldots, g_n$. Let $Y_\ell = g_\ell(X_1, \ldots, X_n)$ for each $\ell \in \mathbb{N}$.
Let $X_1, \ldots, X_n$ be jointly continuous with joint PDF $f_{X_1,\ldots,X_n}$. Then, the joint PDF of $Y_1, \ldots, Y_n$ is given by

$$f_{Y_1,\ldots,Y_n}(y_1, \ldots, y_n) = \sum_{\ell=1}^{n} \frac{f_{X_1,\ldots,X_n}\Big(h_\ell(y_1, \ldots, y_n)\Big)}{\Big|\det J_g\Big(h_\ell(y_1, \ldots, y_n)\Big)\Big|} \mathbf{1}_{g(I_\ell)}(y_1, \ldots, y_n).$$

Problem-01

Suppose that $X, Y \overset{\text{IID}}{\sim} \mathcal{N}(0,1)$. Define R and Θ as

$$R = \sqrt{X^2 + Y^2}, \qquad \Theta = \arctan\frac{Y}{X} = \tan^{-1}\left(\frac{Y}{X}\right).$$

- Determine the joint PDF of R and Θ.
- Determine the marginal PDFs of R and Θ.
- Are R and Θ independent? Justify.

Problem-02

Let R and Θ be jointly continuous with the joint PDF

$$f_{R,\Theta}(r,\theta) \;=\; re^{-r^2/2}\cdot\frac{1}{2\pi}, \qquad r\geq 0, \quad \theta\in[0,2\pi].$$

1. Compute the marginal PDFs of R and Θ, and show that $R \perp\!\!\!\perp \Theta$.
2. Let $X = R\cos(\Theta)$ and $Y = R\sin(\Theta)$. Compute the joint PDF of X and Y.
3. Are X and Y independent? Justify.

Problem-03

Suppose that f denotes the Rayleigh PDF, i.e.,

$$f(r) \;=\; r\,e^{-r^2/2}, \qquad r \geq 0.$$

Let F denote the CDF of the Rayleigh PDF f, and let $g : (0,1) \to \mathbb{R}$ be defined as

$$g(u) \;=\; \min\big\{r :\; F(r) \geq u\big\}, \qquad u \in (0,1).$$

- Give the closed-form expression for g.
- Suppose that $U \sim \text{Uniform}(0,1)$. Determine the PDF of the random variable $X = g(U)$.

Inverse CDF

The function g defined above is the **inverse CDF function** corresponding to the Rayleigh CDF F.

Inverse CDF

Definition (Inverse CDF)

Given a CDF $F : \mathbb{R} \to [0, 1]$, we define its **inverse** $F^{-1} : (0, 1) \to \mathbb{R}$ as

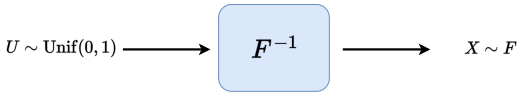
$$F^{-1}(u) \;=\; \min\big\{x \in \mathbb{R} :\; F(x) \geq u\big\}, \qquad u \in (0, 1).$$

- The above inverse function applies even if the CDF is not an invertible function
- If the CDF is invertible, then the above definition of inverse coincides with the usual notion of inverse
- **A universal fact (without proof):**
  Suppose that F is a CDF.
  Let $U \sim \text{Uniform}(0, 1)$, and let $Y = F^{-1}(U)$, where F^{-1} is the inverse CDF as defined above
  Then, the random variable $X = F^{-1}(U)$ will have CDF F, i.e.,

$$F_X \;=\; F.$$

- The above universal fact is the basis for the so-called **inverse transform technique (ITT)** for generating samples from custom distributions on a computer!

Inverse Transform Technique



- Suppose that F is a CDF.
  Let $U \sim \text{Uniform}(0, 1)$, and let $Y = F^{-1}(U)$, where F^{-1} is the inverse CDF
  Then, the random variable $X = F^{-1}(U)$ will have CDF F, i.e.,

$$F_X = F.$$

Gaussian Samples on Python via ITT

Box–Muller Transformation

1. Let $U_1, U_2 \overset{\text{IID}}{\sim} \text{Unif}(0, 1)$
2. Let R and Θ be two random variables defined via

$$R = F^{-1}_{\text{Rayleigh}}(U_1), \qquad \Theta = 2\pi U_2,$$

   where F_{Rayleigh} is the CDF of the Rayleigh distribution
3. Let Y_1 and Y_2 be defined as

$$Y_1 = R\,\cos(\Theta), \qquad Y_2 = R\,\sin(\Theta).$$

4. Then, $Y_1, Y_2 \overset{\text{i.i.d.}}{\sim} \mathcal{N}(0, 1)$
5. To get $X \sim \mathcal{N}(\mu, \sigma^2)$, simply discard Y_2, and set

$$X = \sigma\, Y_1 + \mu.$$

6. Repeat steps 1-5 a total of n times to get $X_1, X_2, \ldots, X_n \overset{\text{i.i.d.}}{\sim} \mathcal{N}(\mu, \sigma^2)$